SIBLINGS OF FM FUNCTORS ARE FM UP TO AUTOMORPHISM-TWIST

THE STACKS PROJECT AUTHORS

Editorial note. Two 'details omitted' spots (sibling quasi-inverses are siblings; the displayed inverse formula is FM) are auxiliary; the formulas are displayed in-span.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 3195, 3197 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Essential image containment. lemma-two-functors-pre L26478 (carded this batch). G lands in the essential image of F.
- Noah classification. lemma-noah L26461 (carded this batch). Writes H as a sibling of f^* tensor L.
- Sibling transfer. lemma-sibling-faithful L26454 (carded this batch). Siblings preserve equivalence and Coh behavior.

Source reference. Original lemma 0FYS: sources/stacks-equiv/pinned-original-source.tex, source lines 1200–1200. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G27: sources/stacks-equiv/pinned-original-source.tex, source lines 3084–3084. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FZV: sources/stacks-equiv/pinned-original-source.tex, source lines 2253–2253. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GX0: sources/stacks-equiv/pinned-original-source.tex, source lines 3039–3039. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let X, Y be proper schemes over k. Assume X regular. Let $F, G : D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_Y)$ be k-linear exact functors such that*

(1) *$F(\mathcal{F}) \cong G(\mathcal{F})$ for any coherent $\mathcal{O}_X$-module $\mathcal{F}$ with $\dim(Supp(\mathcal{F})) = 0$,*
(2) *F is fully faithful, and*
(3) *G is a Fourier-Mukai functor whose kernel is in $D_{perf}(\mathcal{O}_{X \times Y})$.*

Then there exists a Fourier-Mukai functor $F' : D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_Y)$ whose kernel is in $D_{perf}(\mathcal{O}_{X \times Y})$ such that F and F' are siblings.

Proof. The essential image of G is contained in the essential image of F by Lemma 0GX0. Consider the functor $H = F^{-1} \circ G$ which makes sense as F is fully faithful. By Lemma 0G27 we obtain an automorphism $f : X \to X$ and an invertible $\mathcal{O}_X$-module $\mathcal{L}$ such that the functor $H' : K \mapsto f^*K \otimes \mathcal{L}$ is a sibling of H. In particular H is an auto-equivalence by Lemma 0FZV and H induces an auto-equivalence of $Coh(\mathcal{O}_X)$ (as this is true for its sibling functor H'). Thus the quasi-inverses H^{-1} and $(H')^{-1}$ exist, are siblings (small detail omitted), and $(H')^{-1}$ sends M to $(f^{-1})^*(M \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{L}^{\otimes -1})$ which is a Fourier-Mukai functor (details omitted). Then of course $F = G \circ H^{-1}$ is a sibling of $G \circ (H')^{-1}$. Since compositions of Fourier-Mukai functors are Fourier-Mukai by Lemma 0FYS we conclude. □